

Performance Testing

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates the responsiveness, stability, and efficiency of the OptiCrop application during crop prediction. The testing verifies that the Flask-based web application processes user requests quickly, returns accurate crop recommendations, and remains stable during repeated execution. The results demonstrate that the application performs efficiently under normal operating conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Performance Testing, Response Time Testing
Target Module / API	Crop Recommendation Prediction Module (/predict)
Test Environment	Local Development Environment (Flask Application)
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a crop recommendation request with valid soil and weather parameters.	1	30	Crop recommendation generated successfully within the expected response time.
2	Execute multiple crop prediction requests sequentially.	10	60	All requests processed successfully without errors.
3	Perform continuous prediction requests under moderate load.	20	120	Stable response time with consistent prediction results.
4	Submit invalid input values to verify validation and error handling.	5	30	Invalid inputs are rejected with appropriate validation messages.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.72 seconds	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	1.85 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	> 5 Req/sec	12 Req/sec	Pass	Stable request handling
4	Error Rate	< 1%	0%	Pass	No failed requests observed
5	CPU Utilization	< 80%	40%	Pass	Efficient CPU utilization
6	Memory Utilization	< 80%	36%	Pass	Stable memory usage

Step 4: Observations & Analysis

- The OptiCrop application generated crop recommendations quickly under normal operating conditions.
- The trained **Logistic Regression** model produced predictions with low response time.
- No application crashes or failed requests were observed during testing.
- Input validation successfully rejected invalid user inputs.
- CPU and memory utilization remained within acceptable limits throughout execution.
- The application demonstrated stable performance and is suitable for deployment and future enhancements.

Step 5: Screenshots / Evidence

Performance testing was conducted using Postman in the local Flask development environment. Multiple prediction requests were executed to verify response time, prediction accuracy, and application stability.

The application successfully generated crop recommendations for different soil and environmental parameter values without errors. The testing confirmed that the system meets the expected performance requirements for crop recommendation.